



Manajemen Proyek Sistem Informasi

PROJECT SCOPE MANAGEMENT

S1 SISTEM INFORMASI

FAKULTAS REKAYASA INDUSTRI

TELKOM UNIVERSITY





Learning Objectives

- Understand the **importance of good project scope management**
- Describe the **process of planning scope management**
- Discuss methods for **collecting and documenting requirements** in order to meet stakeholder needs and expectations
- Explain the **scope definition process** and describe the contents of a project scope statement
- Discuss the **process for creating a work breakdown structure** using the analogy, top-down, bottom-up, and mind-mapping approaches
- Explain the **importance of verifying scope** and how it relates to defining and controlling scope
- Understand the **importance of controlling scope** and approaches for preventing scope-related problems on information technology projects
- Describe **how software can assist in project scope** management



What is Project Scope Management?

- Scope refers to **all the work involved in creating the products of the project** and the **processes used to create them**
- A deliverable is **a product produced as part of a project**, such as hardware or software, planning documents, or meeting minutes
- Project scope management includes the processes involved in **defining and controlling what is or is not included** in a project



Project Scope Management Processes

1. Planning Processes

- A. **Plan Scope Management**: defining how the scope will be managed
- B. **Collecting requirements**: defining and documenting the features and functions of the products produced during the project as well as the processes used for creating them
- C. **Defining scope**: reviewing the project charter, requirements documents, and organizational process assets to create a scope statement
- D. **Creating the WBS**: subdividing the major project deliverables into smaller, more manageable components

2. Monitoring and Controlling Processes

- A. **Verifying scope**: formalizing acceptance of the project deliverables
- B. **Controlling scope**: controlling changes to project scope throughout the life of the project



Project Scope Management Summary

Planning

Process: **Plan scope management**

Outputs: Scope management plan, requirements management plan

Process: **Collect requirements**

Outputs: Requirements documentation, requirements traceability matrix

Process: **Define scope**

Outputs: Project scope statement, project documents updates

Process: **Create WBS**

Outputs: Scope baseline, project documents updates

Monitoring and Controlling

Process: **Validate scope**

Outputs: Accepted deliverables, change requests, work performance information, project documents updates

Process: **Control scope**

Outputs: Work performance information, change requests, project management plan updates, project documents updates, organizational process assets updates

Project Start

Project Finish



1. Planning Scope Management

- Planning **how the scope will be managed**
- Two important output : **scope management plan** and **requirements management plan**
- Scope management plan is a **subsidiary part of the project management plan** (refer to Project Integration Management)

Planning Scope Management ITTO

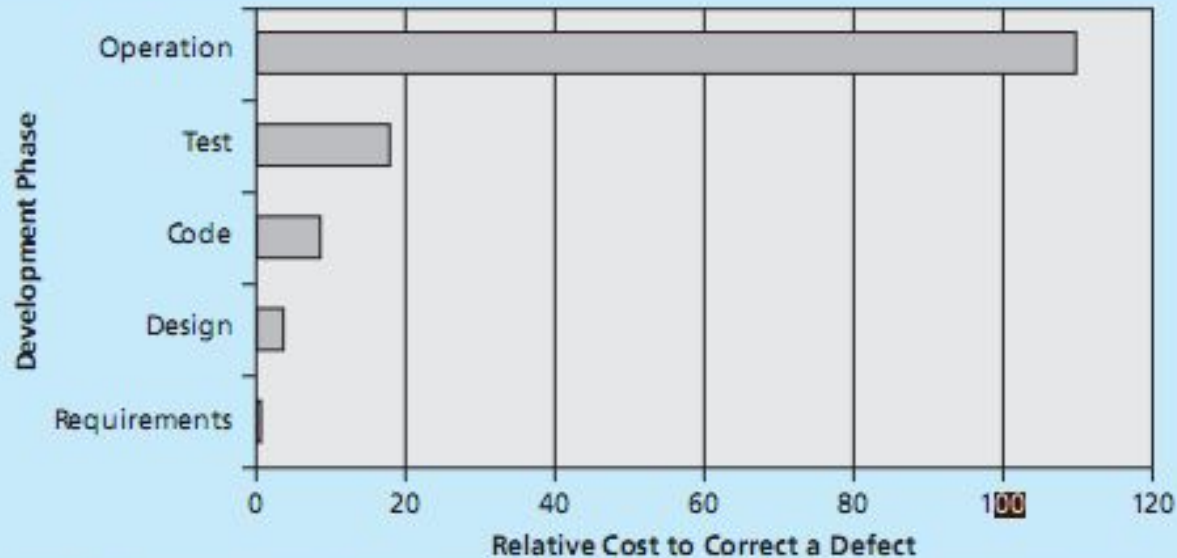




2. Collecting Requirements

- A requirement is “. A condition or capability that is required to be present in a product, service, or result to satisfy a contract or other formally imposed specification.” (PMBOK® Guide, 2013)
- For some IT projects, it is helpful to divide requirements development into categories called elicitation, analysis, specification, and validation
- It is important to use an iterative approach to defining requirements since they are often unclear early in a project

Relative Cost to Correct a Software Requirement Defect



Source: Robert B. Grady, "An Economic Release Decision Model: Insights into Software Project Management." *Proceedings of the Applications of Software Measurement Conference* (Orange Park, FL: Software Quality Engineering, 1999), pp. 227–239.

ITTO Collecting Requirements

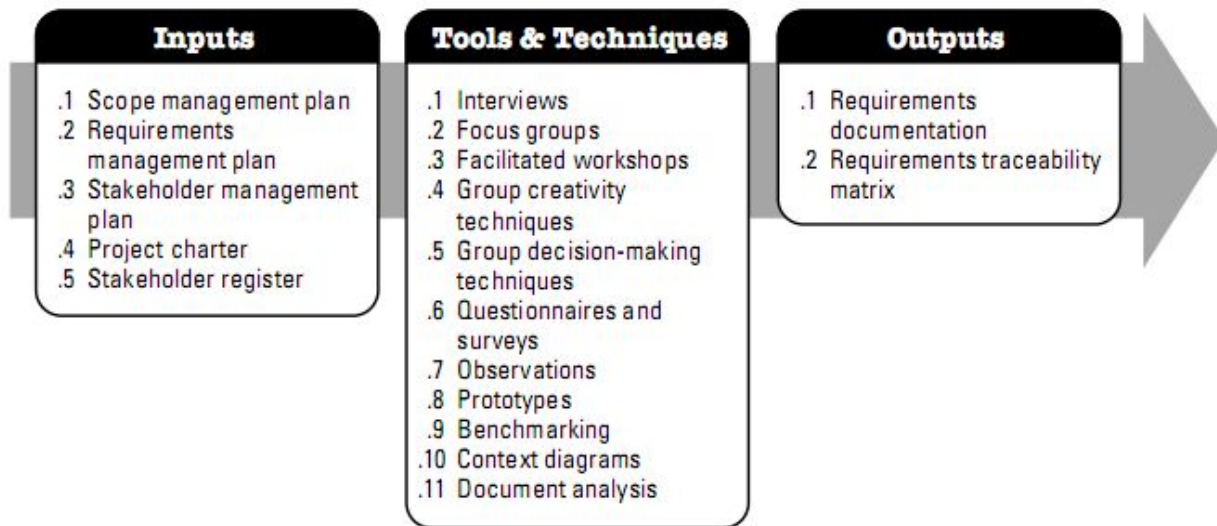


Figure 5-4. Collect Requirements: Inputs, Tools & Techniques, and Outputs



Methods for Collecting Requirements

- Interviewing
- Focus groups and facilitated workshops
- Using group creativity and decision-making techniques
- Questionnaires and surveys
- Observation
- Prototyping
- Software tools





What Went Right?

- Genesys Telecommunications Laboratories uses **Accept software**, a **product planning and innovation management application** and winner of the Excellence in Product Management Award from 2006–2008
- Accept helps them instill a **consistent, repeatable**, and **predictable** process for **new product definition** and development
- They can define what **information comprises a requirement** and enforce discipline around that process



Documenting Requirements

- Requirements documents are often generated by software and include text, images, diagrams, videos, and other media; they are often broken down into different categories such as functional, service, performance, quality, training requirements, and so on
- A requirements management plan describes how project requirements will be analyzed, documented, and managed
- A requirements traceability matrix (RTM) is a table that lists requirements, various attributes of each requirement, and the status of the requirements to ensure that all requirements are addressed



Sample Requirements Traceability Matrix

Requirement No.	Name	Category	Source	Status
R32	Laptop memory	Hardware	Project charter and corporate laptop specifications	Complete. Laptops ordered meet requirement by having 4GB of memory.

3. Defining Scope & ITTO

- Define scope is the process of developing a detailed description of the project and product



Figure 5-7. Define Scope: Inputs, Tools & Techniques, and Outputs



Defining Scope (Cont)

- Key inputs for preparing the project scope statement include the project charter, requirements documentation, and organizational process assets such as policies and procedures related to scope statements as well as project files and lessons learned from previous, similar projects
- As time progresses, the scope of a project should become more clear and specific



Sample Project Charter

Project Title: Information Technology (IT) Upgrade Project

Project Start Date: March 4

Projected Finish Date: December 4

Key Schedule Milestones:

- Inventory update completed April 15
- Hardware and software acquired August 1
- Installation completed October 1
- Testing completed November 15

Budget Information: Budgeted \$1,000,000 for hardware and software costs and \$500,000 for labor costs.

Project Manager: Kim Nguyen, (310) 555-2784, knguyen@course.com





Sample Project Charter (Cont)

Project Objectives: Upgrade hardware and software for all employees (approximately 2,000) within nine months based on new corporate standards. See attached sheet describing the new standards. Upgrades may affect servers, as well as associated network hardware and software.

Main Project Success Criteria: The hardware, software, and network upgrades must meet all written specifications, be thoroughly tested, and be completed in less than ten months. Employee work disruptions will be minimal.

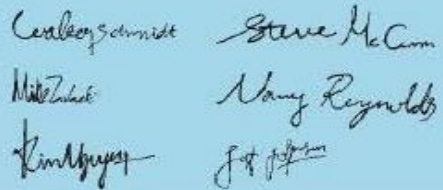
Approach:

- Update the information technology inventory database to determine upgrade needs
- Develop detailed cost estimate for project and report to CIO
- Issue a request for quote to obtain hardware and software
- Use internal staff as much as possible for planning, analysis, and installation





Sample Project Charter (Cont)

ROLES AND RESPONSIBILITIES		
Name	Role	Responsibility
Walter Schmidt	CEO	Project sponsor, monitor project
Mike Zwack	CIO	Monitor project, provide staff
Kim Nguyen	Project Manager	Plan and execute project
Jeff Johnson	Director of Information Technology Operations	Mentor Kim
Nancy Reynolds	VP, Human Resources	Provide staff, issue memo to all employees about project
Steve McCann	Director of Purchasing	Assist in purchasing hardware and software
Sign-off: (Signatures of all the above stakeholders)		
		
Comments: (Handwritten or typed comments from above stakeholders, if applicable)		
“This project must be done within ten months at the absolute latest.” Mike Zwack, CIO		
“We are assuming that adequate staff will be available and committed to supporting this project. Some work must be done after hours to avoid work disruptions, and overtime will be provided.” Jeff Johnson and Kim Nguyen, Information Technology department.		



Further Defining Project Scope

Project Charter:

Upgrades may affect servers . . . (listed under Project Objectives)

Project Scope Statement, Version 1:

Servers: If additional servers are required to support this project, they must be compatible with existing servers. If it is more economical to enhance existing servers, a detailed description of enhancements must be submitted to the CIO for approval. See current server specifications provided in Attachment 6. The CEO must approve a detailed plan describing the servers and their location at least two weeks before installation.

Project Scope Statement, Version 2:

Servers: This project will require purchasing ten new servers to support Web, network, database, application, and printing functions. Virtualization will be used to maximize efficiency. Detailed descriptions of the servers are provided in a product brochure in Appendix 8 along with a plan describing where they will be located.





4. Creating the Work Breakdown Structure (WBS)

- A WBS is a deliverable-oriented grouping of the work involved in a project that defines the total scope of the project
- WBS is a foundation document that provides the basis for planning and managing project schedules, costs, resources, and changes
- Decomposition is subdividing project deliverables into smaller pieces
- A work package is a task at the lowest level of the WBS

Creating the Work Breakdown Structure (WBS) ITTO

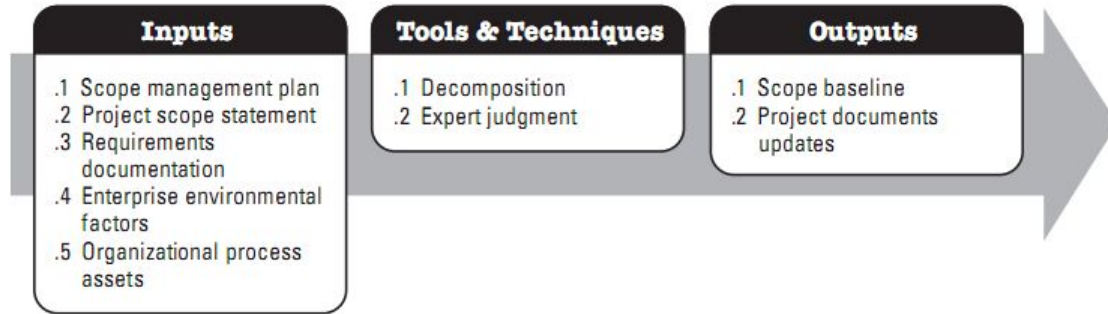
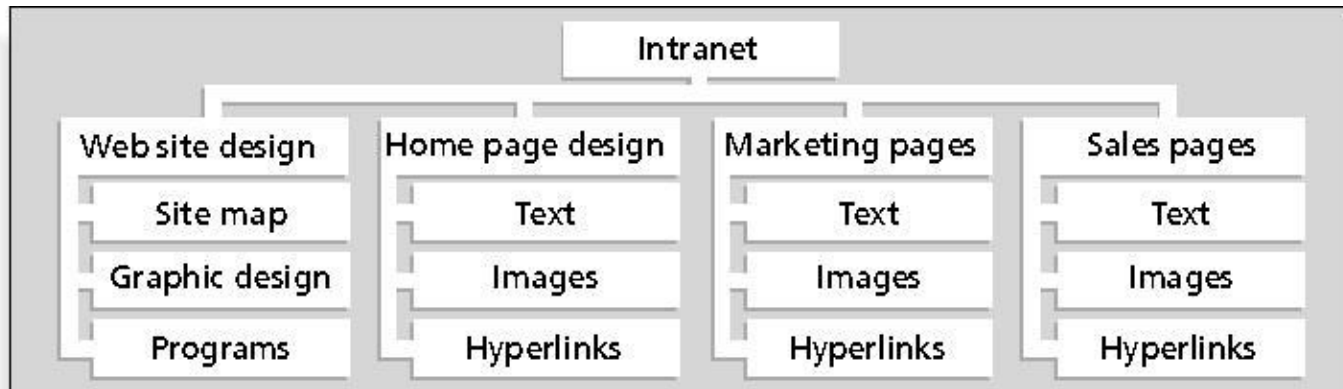


Figure 5-9. Create WBS: Inputs, Tools & Techniques, and Outputs

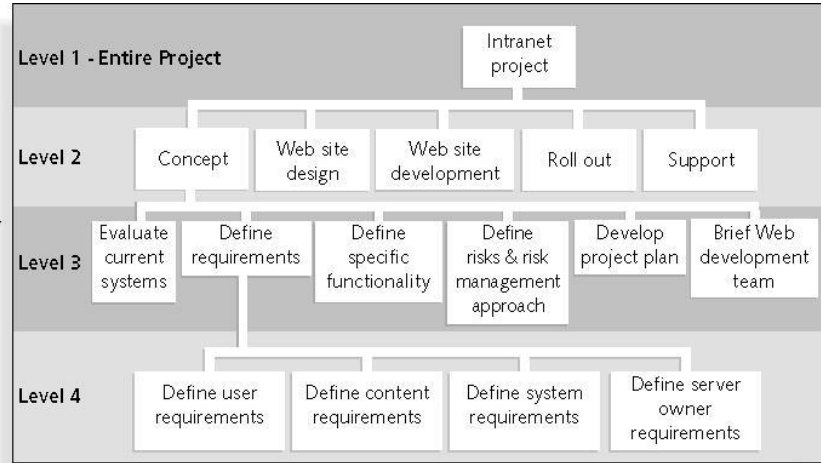
Sample Intranet WBS Organized by Product





Sample Intranet WBS Organized by Phase

Chart
form →



Tabular form with Microsoft Project numbering

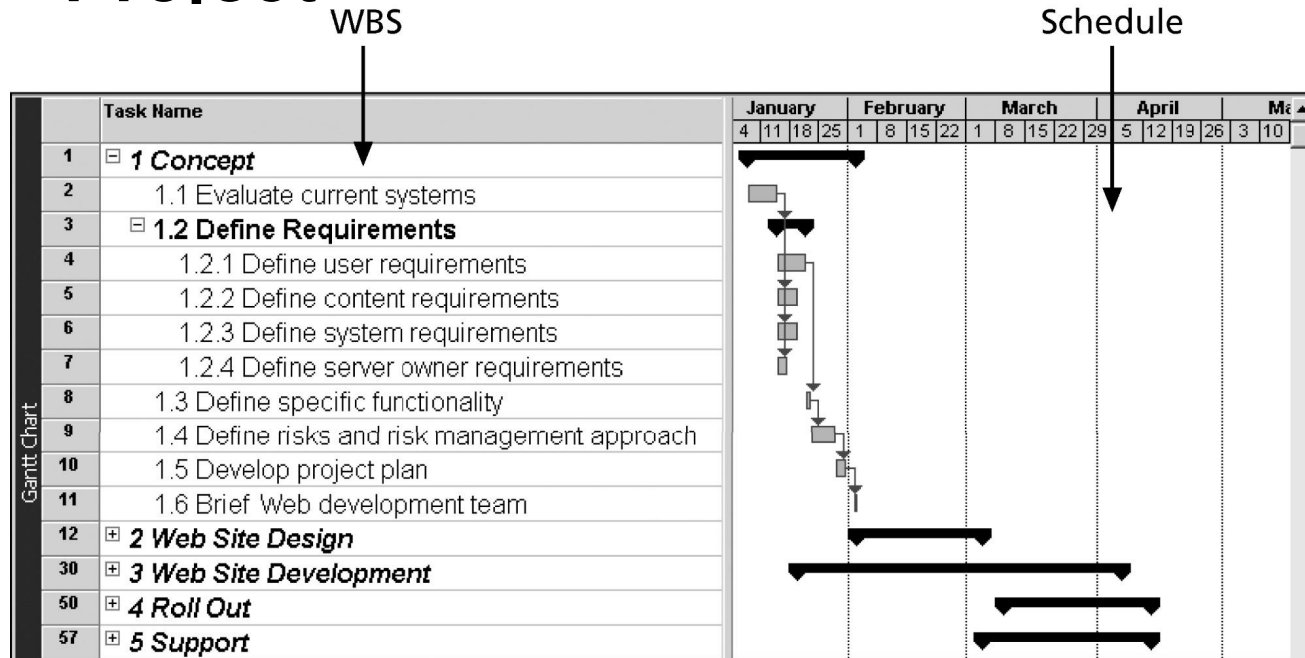
- 1.0 Concept
 - 1.1 Evaluate current systems
 - 1.2 Define requirements
 - 1.2.1 Define user requirements
 - 1.2.2 Define content requirements
 - 1.2.3 Define system requirements
 - 1.2.4 Define server owner requirements
 - 1.3 Define specific functionality
 - 1.4 Define risks and risk management approach
 - 1.5 Develop project plan
 - 1.6 Brief Web development team
- 2.0 Web site design
- 3.0 Web site development
- 4.0 Roll out
- 5.0 Support

Tabular form with PMI numbering

- 1.1 Concept
 - 1.1.1 Evaluate current systems
 - 1.1.2 Define requirements
 - 1.1.2.1 Define user requirements
 - 1.1.2.2 Define content requirements
 - 1.1.2.3 Define system requirements
 - 1.1.2.4 Define server owner requirements
 - 1.1.3 Define specific functionality
 - 1.1.4 Define risks and risk management approach
 - 1.1.5 Develop project plan
 - 1.1.6 Brief Web development team
- 1.2 Web site design
- 1.3 Web site development
- 1.4 Roll out
- 1.5 Support



Intranet **WBS and Gantt Chart** in Microsoft Project

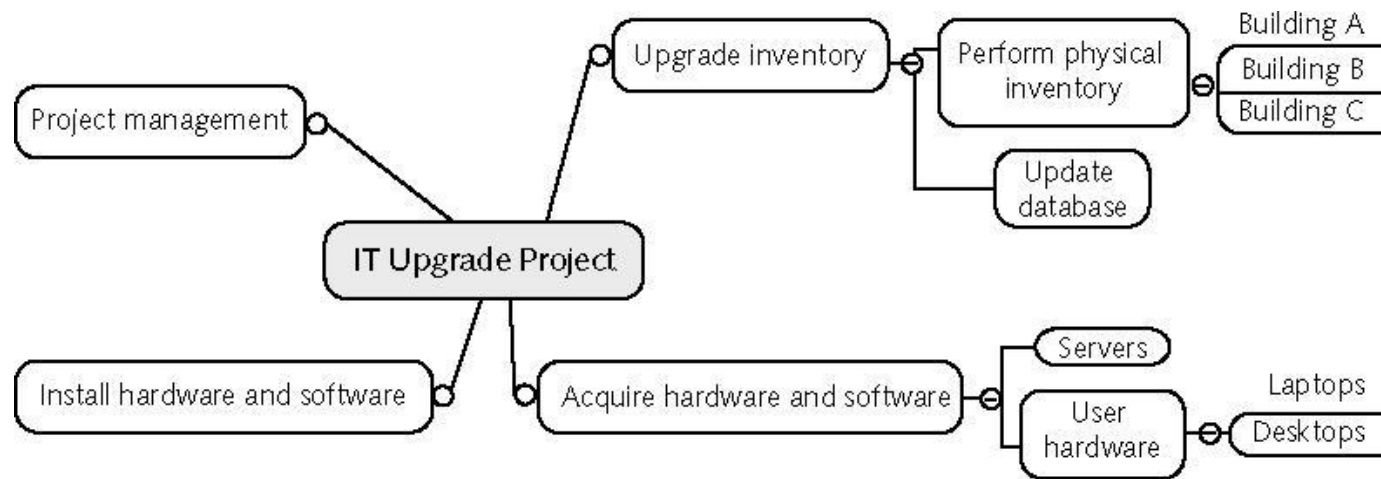




Approaches to Developing WBSs

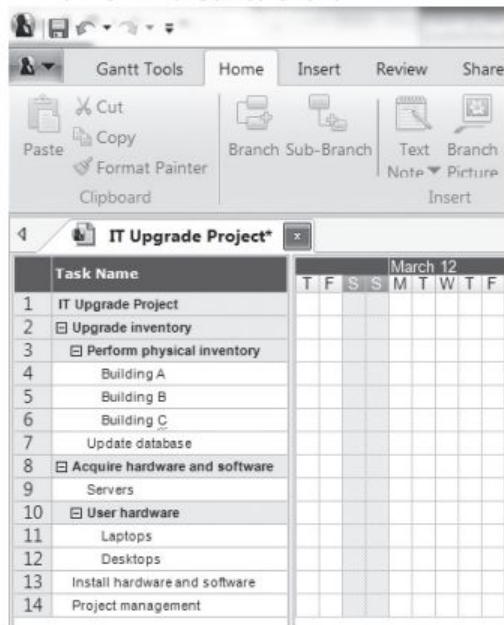
1. **Using guidelines:** some organizations, like the DOD, provide **guidelines for preparing WBSs**
2. **The analogy approach:** review WBSs of similar projects and tailor to your project
3. **The top-down approach:** start with the largest items of the project and break them down
4. **The bottom-up approach:** start with the specific tasks and roll them up
5. **Mind-mapping approach:** mind mapping is a technique that uses **branches radiating out from a core idea** to structure thoughts and ideas

Sample Mind-Mapping Approach for Creating a WBS

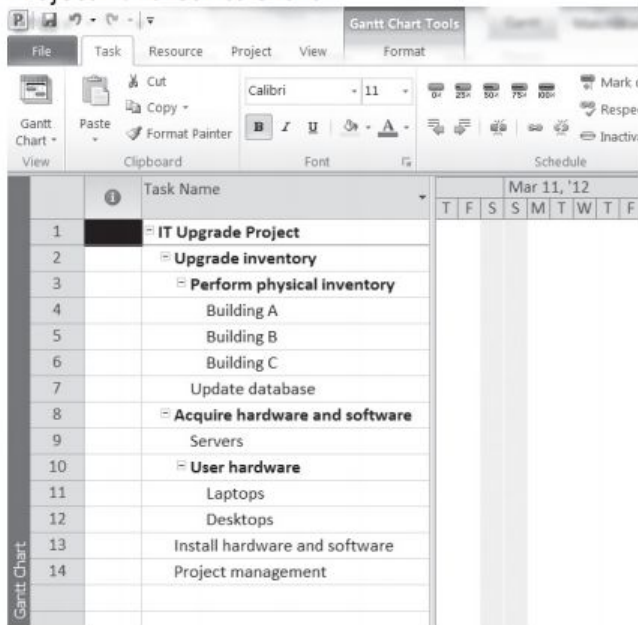


WBS Generated from a Mind Mapping

MindView 4.0 Gantt Chart



Project 2010 Gantt Chart





The WBS Dictionary and Scope Baseline

- Many WBS tasks are vague and must be explained more so people know what to do and can estimate how long it will take and what it will cost to do the work
- A WBS dictionary is a document that describes detailed information about each WBS item
- The approved project scope statement and its WBS and WBS dictionary form the scope baseline, which is used to measure performance in meeting project scope goals



WBS Dictionary Example

WBS Dictionary Entry March 20

Project Title: Information Technology (IT) Upgrade Project

WBS Item Number: 2.2

WBS Item Name: Update Database

Description: The IT department maintains an online database of hardware and software on the corporate Intranet. However, we need to make sure that we know exactly what hardware and software employees are currently using and if they have any unique needs before we decide what to order for the upgrade. This task will involve reviewing information from the current database, producing reports that list each department's employees and location, and updating the data after performing the physical inventory and receiving inputs from department managers. Our project sponsor will send out a notice to all department managers to communicate the importance of this project and this particular task. In addition to general hardware and software upgrades, the project sponsors will ask the department managers to provide information for any unique requirements they might have that could affect the upgrades. This task also includes updating the inventory data for network hardware and software. After updating the inventory database, we will send an e-mail to each department manager to verify the information and make changes online, as needed. Department managers will be responsible for ensuring that their people are available and cooperative during the physical inventory. Completing this task is dependent on WBS Item Number 2.1, Perform Physical Inventory and must precede WBS Item Number 3.0, Acquire Hardware and Software.



Advice for Creating a WBS and WBS Dictionary

- A **unit of work should appear at only one place** in the WBS
- The work content of a WBS item is the sum of the WBS items below it
- A **WBS item is the responsibility of only one individual**, even though many people may be working on it
- The WBS **must be consistent** with the way in which work is actually going to be performed; it should serve the project team first and other purposes only if practical



Advice for Creating a WBS and WBS Dictionary (continued)

- Project team members should be involved in developing the WBS to ensure consistency and buy-in
- Each WBS item must be documented in a WBS dictionary to ensure accurate understanding of the scope of work included and not included in that item
- The WBS must be a flexible tool to accommodate inevitable changes while properly maintaining control of the work content in the project according to the scope statement



What Went Wrong?

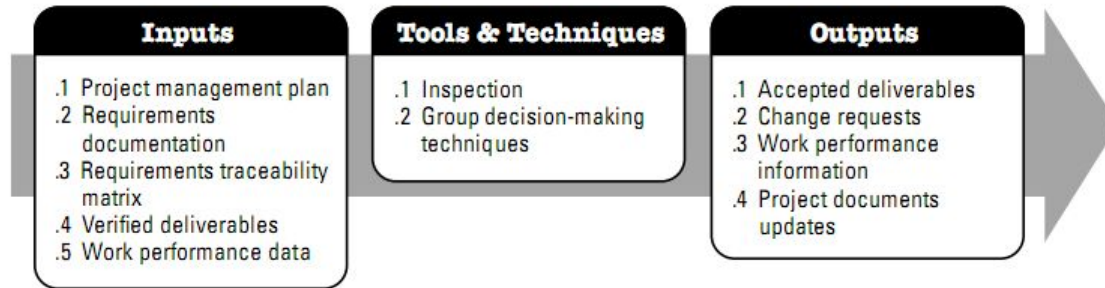
- A project scope that is too broad and grandiose can cause severe problems
 - **Scope creep** and an overemphasis on technology for technology's sake **resulted in the bankruptcy of a large pharmaceutical firm, Texas-based FoxMeyer Drug**
 - In 2001, McDonald's fast-food chain initiated a project to create an intranet that would connect its headquarters with all of its restaurants to provide detailed operational information in real time; after spending \$170 million on consultants and initial implementation planning, McDonald's realized that the **project was too much to handle and terminated it**



5. Validate Scope

- It is **very difficult to create a good scope statement** and WBS for a project
- It is even **more difficult to verify project scope and minimize scope changes**
- **Scope verification** involves **formal acceptance of the completed project scope by the stakeholders**
- **Acceptance** is often **achieved by a customer inspection and then sign-off on key deliverables**

Validate Scope ITTO





6. Controlling Scope

- Scope control involves **controlling changes to the project scope**
- **Goals of scope control** are to:
 - **Influence the factors** that cause scope changes
 - **Assure changes** are processed according to procedures developed as part of integrated change control
 - **Manage changes** when they occur
- **Variance** is the **difference between planned and actual performance**
- **Variance Analysis is the only tools in controlling scope**

Controlling Scope ITTO

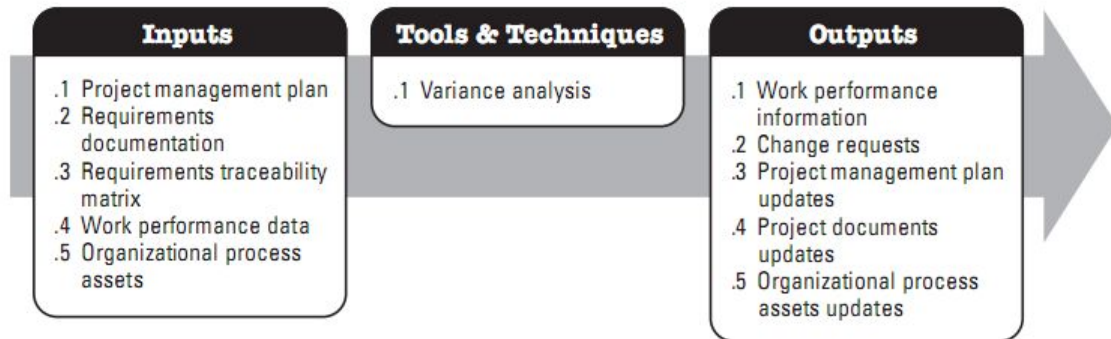


Figure 5-16. Control Scope: Inputs, Tools & Techniques, and Outputs



Best Practices for Avoiding Scope Problems

1. **Keep the scope realistic.** Don't make projects so large that they can't be completed. Break large projects down into a series of smaller ones.
2. **Involve users in project scope management.** Assign key users to the project team and give them ownership of requirements definition and scope verification.
3. **Use off-the-shelf hardware and software whenever possible.** Many IT people enjoy using the latest and greatest technology, but business needs, not technology trends, must take priority.
4. **Follow good project management processes.** As described in this chapter and others, there are well-defined processes for managing project scope and others aspects of projects.



Suggestions for Improving User Input

- **Develop a good project selection process** and insist that sponsors are from the user organization
- **Have users** on the project team in important roles
- **Have regular meetings** with defined agendas, and have users sign off on key deliverables presented at meetings
- **Deliver something to users** and sponsors on a regular basis
- **Don't promise to deliver** when you know you can't
- **Co-locate users with developers**



Suggestions for Reducing Incomplete and Changing Requirements

- Develop and **follow a requirements management process**
- **Use techniques such as prototyping**, use case modeling, and JAD to get more user involvement
- **Put requirements in writing** and keep them current
- **Create a requirements management database** for documenting and controlling requirements



Suggestions for Reducing Incomplete and Changing Requirements (cont.)

- Provide **adequate testing** and conduct testing throughout the project life cycle
- **Review changes** from a systems perspective
- **Emphasize completion dates** to help focus on what's most important
- **Allocate resources specifically for handling change** requests/enhancements like NWA did with ResNet



Using Software to Assist in Project Scope Management

- **Word-processing software** helps create several scope-related documents
- **Spreadsheets** help to perform financial calculations and weighted scoring models and to develop charts and graphs
- Communication software like **e-mail and the Web** help clarify and communicate scope information
- **Project management software** helps in creating a WBS, the basis for tasks on a Gantt chart
- Specialized software is available to assist in project scope management



Summary

- Project scope management includes the processes required to ensure that the project addresses all the work required, and only the work required, to complete the project successfully
- Main processes include:
 1. Create Scope Management Plan
 2. Collect requirements
 3. Define scope
 4. Create WBS
 5. Verify scope
 6. Control scope



References

1. Kathy Schwalbe, **Managing Information Technology Projects 7th Edition**, *Course Technology, Cengage Learning*, 2012
2. A Guide to the Project Management Body of Knowledge: **PMBOK Guide 6th Edition**, *Project Management Institute*, 2017